

BEYOND THE PAGE



The momentum factor

www.mhhe.com/brealey11e

1931, there would have been very little relation between their portfolio returns and beta.⁸ Does this imply that there has been a fundamental change in the relation between risk and return in the last 51 years, or did high-beta stocks just happen to perform worse during these years than investors initially expected? It is hard to be sure.

There is little doubt that the CAPM is too simple to capture everything that is going on in the market. For example, look at Figure 12.8. The orange line shows the cumulative difference between the returns on small-firm stocks and large-firm stocks. If you had bought the shares with the smallest market capitalizations and sold those with the largest capitalizations, this is how your wealth would have changed. You can see that small-cap stocks did not always do well, but over the long haul, their owners have made substantially higher returns. Since the end of 1926, the average annual difference between the returns on the two groups of stocks has been 2.4%. Now look at the blue line in Figure 12.8, which shows the cumulative difference between the returns on value stocks and growth stocks. *Value stocks* here are defined as those with high ratios of book value to market value. *Growth stocks* are those with low ratios of book to market. Notice that value stocks have provided a higher long-run return than growth stocks. Since 1926, the average annual difference between returns on value and growth stocks has been about 3.9%.

The superior performance of small-firm stocks and value stocks does not fit well with the CAPM, which predicts that beta is the only reason that expected returns differ. If investors expected the returns to depend on firm size or book-to-market ratios, then the simple version of the capital asset pricing model cannot be the whole truth.

What's going on here? It is hard to say. Defenders of the capital asset pricing model emphasize that it is concerned with *expected* returns, whereas we can observe only *actual* returns. Actual returns reflect expectations, but they also embody lots of "noise"—the steady flow of surprises that conceal whether on average investors have received the returns that they expected. Thus, when we observe that in the past small-firm stocks and value stocks have provided superior performance, we can't be sure whether this was simply a coincidence or whether investors have required a higher expected return to hold these stocks.

Such debates have prompted headlines like "Is Beta Dead?" in the business press. It is not the first time that beta has been declared dead, but the CAPM remains the leading model for estimating required returns. Only strong theories can have more than one funeral.

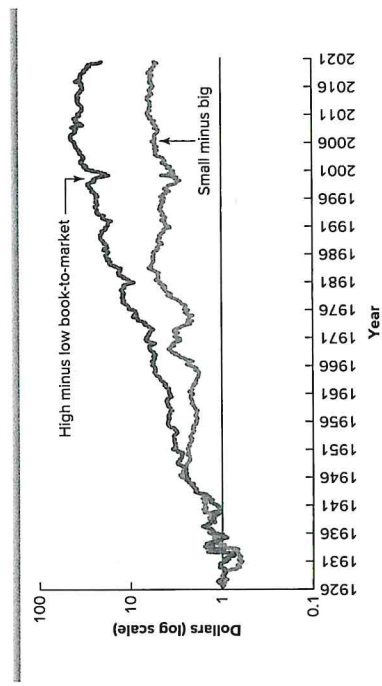


FIGURE 12.8 The orange line shows the cumulative difference between the returns on small-firm and large-firm stocks from 1926 to 2020. The blue line shows the cumulative difference between the returns on high-book-to-market-value stocks and low-book-to-market-value stocks.

Source: mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

⁸ During this later period, the returns to the first seven investors increased in line with beta. However, the highest-beta portfolios performed poorly.

The CAPM is not the only model of risk and return. It has several brothers and sisters as well as second cousins. However, the CAPM captures in a simple way two fundamental ideas. First, almost everyone agrees that investors require some extra return for taking on risk. Second, investors appear to be concerned principally with the market risk that they cannot eliminate by diversification. That is why nearly three-quarters of financial managers use the capital asset pricing model to estimate the cost of capital.⁹

The CAPM and the Opportunity Cost of Capital

The discount rate for valuing a proposed capital investment project should be the *opportunity cost of capital*, defined as the expected rate of return that the company's shareholders could achieve by investing on their own. But the CAPM tells us (confirming common sense) that expected rates of return depend on risk, that is, on beta. Therefore, the opportunity cost of capital for a proposed project should depend on the project's beta. The **project cost of capital** is, therefore, its minimum acceptable expected rate of return, given its risk.

12.4

project cost of capital
Minimum acceptable
expected rate of return on
a project given its risk.

Example 12.4

Estimating the Opportunity Cost of Capital for a Project

Suppose that Coca-Cola is contemplating an investment of \$10 million to expand a warehouse. It has forecast cash flows and calculated an internal rate of return (IRR) of 3%. We assume, as in Table 12.2, that the risk-free rate is 1% and the market risk premium is 7%. Should Coke invest?

To answer this question, you need to know the opportunity cost of capital r , defined as the expected rate of return that Coke's shareholders could achieve if the \$10 million were paid out to them to invest on their own. The CAPM says that the expected rate of return demanded by investors depends on the project beta.

Suppose that the project's cash flows are an absolutely sure thing. Then, $\beta = 0$ and the project cost of capital is

$$r = r_f + \beta \times (r_m - r_f) = 1\% + 0 \times 7\% = 1\%$$

If the project offers a 3% return when the opportunity cost of capital for the project is 1%, Coke should obviously go ahead.¹⁰

What about the 4.0% rate of return for Coke in Table 12.2? It's irrelevant if we are confident that the project is risk-free. Coke's stock offers more than the risk-free rate because it is risky, with a beta of .43. Coke's stockholders would all vote for Coke to invest on their behalf in a risk-free project earning 3% because the stockholders could get only 1% investing risk-free on their own.

Surefire projects rarely occur outside finance textbooks. So let's think about the opportunity cost of capital if the project has the same risk as the market portfolio. In this case, $\beta = 1.0$, and the cost of capital is the expected rate of return on the market.

$$r = r_f + \beta \times (r_m - r_f) = 1\% + 1.0 \times 7\% = 8\%$$

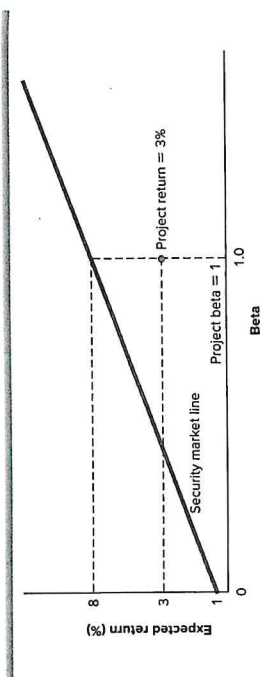
Now that the project beta is 1.0, the project is no longer worth doing. The stockholders would vote against the project at a 3% expected return because they could earn a higher return, 8%, at the same level of risk (same beta) by investing on their own in the market portfolio.

When its beta is 1.0, the project is no longer attractive because, as Figure 12.9 shows, its expected rate of return now lies below the security market line. The project offers a lower rate of return than investors can expect on equally risky investments. Therefore, it is negative-NPV. ■

⁹ See J. R. Graham and C. R. Harvey, "The Theory and Practice of Corporate Finance: Evidence from the Field," *Journal of Financial Economics* 60 (2001), pp. 187–243. A number of managers surveyed reported using more than one method to estimate the cost of capital. Seventy-three percent used the CAPM, while 39% stated they used the average historical stock return, and 34% used the CAPM with additional risk factors.

¹⁰ In Chapter 8, we described some special cases where you should prefer projects that offer a lower internal rate of return than the cost of capital. We assume here that your project is a "normal" one and that you prefer high IRRs to low ones.

FIGURE 12.9 The expected return of this project is less than the expected return one could earn on stock market investments with the same market risk (beta). Therefore, the project's expected return lies below the security market line and the project should be rejected.



If the CAPM holds, the security market line defines the opportunity cost of capital. If a project's expected rate of return plots above the security market line, then it offers a higher expected rate of return than investors could get on their own at the same beta.

The Company Cost of Capital

The cost of capital depends on the use to which the capital is put. It depends on the risk (beta) of the project—not on the risk of the company investing in the project. If a company has a low-risk project, it should discount project cash flows at a correspondingly low rate. If it has a high-risk project, it should discount at a correspondingly high rate. That is the principle. But you can imagine the chaos in practice if a large company had to set a different discount rate for every one of thousands of investment projects. Therefore, most companies estimate a **company cost of capital**, which depends on the average risk of its investments. Many companies use the company cost of capital for all capital-investment projects, which is fine provided that all projects are close enough to average risk. Others set the company cost of capital as a benchmark and adjust the discount rate up or down for riskier or safer projects.

Some companies set two or more discount rates for different types of projects. For example, several electric utilities have both regulated and "merchant" businesses. The regulated businesses are low-risk because they are not allowed to earn much more than a set rate of return. Their profit upside is limited, but so is the downside because they can pass most costs through to their customers and can petition to increase prices if their profits sag. (PG&E, which has a beta of only .55 (see Table 12.2), is mostly a regulated business.) Merchant electricity production, on the other hand, is unregulated and must sell power at fluctuating market prices. Merchant producers can make or lose a lot of money, depending on uncertain demand and costs of production. Companies with both regulated and merchant businesses typically set two costs of capital, one for regulated investments and one for merchant investments.

12.7 Self-Test

Suppose that Walmart is planning an Internet subsidiary to compete head-to-head with Amazon. It argues that it can compete aggressively in Internet retailing because it has a low beta and its shareholders will be content with an expected rate of return of only 3.5%. Is the argument correct?

Many U.S. corporations use the CAPM to compute their company costs of capital. Expected rates of return like those computed for Table 12.2 are only the first step, however. This is because they are *costs of equity*, that is, expected rates of return on common stock. But most companies use debt as well as equity financing. They calculate the company cost of capital as a weighted average of the cost of debt and the cost of equity. We explain how this is done in the next chapter.

What Determines Project Risk?

We have seen that the company cost of capital is the correct discount rate for projects that have the same risk as the company's existing business but *not* for those projects that are safer or riskier than the company's average. How do we know whether a project is unusually risky? Estimating project risk is never going to be an exact science, but here are two things to bear in mind.

First, we saw in Chapter 10 that operating leverage increases the risk of a project. When a large fraction of your costs is fixed, any change in revenues can have a dramatic effect on earnings. Therefore, projects that involve high fixed costs tend to have higher betas.

Second, many people intuitively associate risk with the variability of earnings. But much of this variability reflects diversifiable risk. Lone prospectors in search of gold look forward to extremely uncertain future earnings, but whether they strike it rich is not likely to depend on the performance of the rest of the economy. These investments (like Newmont Mining) have a high standard deviation but a low beta.

What matters is the strength of the relationship between the firm's earnings and the aggregate earnings of all firms. Cyclical businesses, whose revenues and earnings are strongly dependent on the state of the economy, tend to have high betas and a high cost of capital. By contrast, businesses that produce essentials, such as food, beer, and cosmetics, are less affected by the state of the economy. They tend to have low betas and a low cost of capital.

Don't Add Fudge Factors to Discount Rates

Risk to an investor arises because an investment adds to the spread of possible portfolio returns. To a diversified investor, risk is predominantly market risk. But in everyday usage *risk* simply means "bad outcome." People think of the "risks" of a project as the things that can go wrong. For example,

- A geologist looking for oil worries about the risk of a dry hole.
- A pharmaceutical manufacturer worries about the risk that a new vaccine may not be approved by the Food and Drug Administration.
- The owner of a hotel in a politically unstable part of the world worries about the political risk of expropriation.

Managers sometimes add fudge factors to discount rates to account for worries such as these.

This sort of adjustment makes us nervous. First, the bad outcomes we cited appear to reflect diversifiable risks that would not affect the expected rate of return demanded by investors. Second, the need for an adjustment in the discount rate usually arises because managers fail to give bad outcomes their due weight in cash-flow forecasts. They then try to offset that mistake by adding a fudge factor to the discount rate. For example, if a manager is worried about the possibility of a bad outcome such as a dry hole in oil exploration, he or she may reduce the value of the project by using a higher discount rate. That's not the way to do it. Instead, the possibility of the dry hole should be included in the calculation of the expected cash flows to be derived from the well. Suppose that there is a 50% chance of a dry hole and a 50% chance that the well will produce oil worth \$20 million. Then the *expected* cash flow is not \$20 million but $(.5 \times 0) + (.5 \times 20) = \10 million. You should discount the \$10 million expected cash flow at the opportunity cost of capital; it does not make sense to discount the \$20 million using a fudged discount rate.

Expected cash-flow forecasts should already reflect the probabilities of all possible outcomes, good and bad. If the cash-flow forecasts are prepared properly, the discount rate should reflect only the market risk of the project. It should not be fudged to offset errors or biases in the cash-flow forecast.

SUMMARY

How can you measure and interpret the market risk, or beta, of a security? (LO12-1)

- The sensitivity of a stock to market movements is known as **beta**. Stocks with a beta greater than 1.0 are particularly sensitive to market fluctuations. Those with a beta of less than 1.0 are not so sensitive to such movements. The average beta of all stocks is 1.0.
- For diversified investors, the contribution of an investment to the risk of the overall portfolio depends on beta, not total variability.

What is the relationship between the market risk of a security and the rate of return that investors demand of that security? (LO12-2)

- Because high beta stocks contribute more to the variability of diversified portfolios than low beta stocks, they command a high-risk premium.
- The **market risk premium**—that is, the risk premium on the market portfolio—provides a benchmark risk premium for an average risk investment—that is, an investment with a beta of 1. The risk premium on the market portfolio averaged 7.8% between 1900 and 2020. Riskier investments, with betas greater than 1, should provide greater risk premiums than the market, and low-beta investments should offer lower risk premiums.
- The **capital asset pricing model** states that the expected risk premium of an investment should equal the product of its beta and the market risk premium. The expected rate of return from any investment is equal to the risk-free interest rate plus the investment's risk premium, so the CAPM boils down to

$$r = r_f + \beta(r_m - r_f)$$

- The **security market line** is the graphical representation of the CAPM equation. The security market line shows how the return that investors demand is related to the security's beta.

What determines the opportunity cost of capital for a project? (LO12-3)

- The opportunity cost of capital is the return that investors give up by investing in the project rather than in securities of equivalent risk.
- The CAPM implies that the opportunity cost of capital depends on the project's beta. In contrast, the company cost of capital is the expected rate of return demanded by investors in a company. It depends on the average risk of the entirety of the company's assets and operations.
- The opportunity cost of capital for a *specific* project is determined by the use to which the capital is put. Therefore, required rates of return depend on the risk of the project, not on the risk of the firm's existing business. The **project cost of capital** is the minimum acceptable expected rate of return on a project given its risk.
- Your cash-flow forecasts should already factor in the chances of pleasant and unpleasant surprises. Potential bad outcomes should be reflected in the discount rate only to the extent that they affect beta.

LISTING OF EQUATIONS

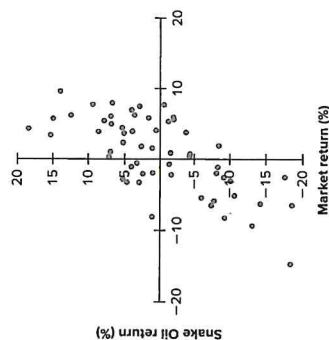
- 12.1 Beta of portfolio = (fraction of portfolio in stock 1 $\times$ beta of stock 1)
+ (fraction of portfolio in stock 2 $\times$ beta of stock 2)
- 12.2 Expected return = risk-free rate + expected risk premium
 $r = r_f + \beta(r_m - r_f)$

QUESTIONS AND PROBLEMS



- Diversifiable Risk.** In light of what you've learned about market versus diversifiable (specific) risks, explain why an insurance company has no problem in selling life insurance to individuals but is reluctant to issue policies insuring against flood damage to residents of coastal areas. Why don't the insurance companies simply charge coastal residents a premium that reflects the actuarial probability of damage from hurricanes and other storms? (LO12-1)
- Specific versus Market Risk.** Figure 12.10 plots monthly rates of return from 2017 to 2021 for the Snake Oil mutual fund. Was this fund fully diversified? (LO12-1)
- Using Beta.** A stock with a beta of .8 has an expected rate of return of 12%. If the market return this year turns out to be 5 percentage points below expectations, what is your best guess as to the rate of return on the stock? (LO12-1)
- Specific versus Market Risk.** Figure 12.11 shows plots of monthly rates of return on three stocks versus the stock market index. (The plots are similar to those in Figure 12.2.) The beta and standard deviation of each stock are given beside its plot. (LO12-1)
 - Which stock is safest for a diversified investor?
 - Which stock is safest for an undiversified investor who puts all her funds in one of these stocks?
 - Consider a portfolio with equal investments in each stock. What would this portfolio's beta have been?
 - Consider a well-diversified portfolio made up of stocks with the same beta as Intel. What are the beta and standard deviation of this portfolio's return? Assume the standard deviation of the market portfolio's return is 20%.
 - What is the expected rate of return on each stock? Use the capital asset pricing model with a market risk premium of 8% and a risk-free rate of interest of 4%.

FIGURE 12.10 Monthly rates of return for the Snake Oil mutual fund and the Standard & Poor's Composite Index.



- Beta.** Go to Connect and link to the material for Chapter 12, where you will find a spreadsheet containing five years of monthly prices for Starbucks (SBUX), U.S. Steel (X), and the S&P 500. Calculate the monthly price changes. (LO12-1)
 - Calculate the beta of each firm. Use Excel's SLOPE function, which fits a regression line through a scatter diagram of two series of numbers.
 - Does the relative magnitude of each beta make sense in terms of the business risk of the two firms? Explain.

The Weighted-Average Cost of Capital and Company Valuation

LEARNING OBJECTIVES

After studying this chapter, you should be able to:

- 13-1 Calculate the weighted-average cost of capital.
- 13-2 Understand when the weighted-average cost of capital is—or isn't—the appropriate discount rate for a new project.
- 13-3 Measure a company's capital structure.
- 13-4 Estimate the expected returns on a firm's securities.
- 13-5 Use the weighted-average cost of capital to value a business given forecasts of its future cash flows.

RELATED WEBSITES FOR THIS CHAPTER CAN BE FOUND IN CONNECT.



Geothermal Corporation was founded to produce electricity from geothermal energy trapped under the earth. How should Geothermal determine its cost of capital? www.jodymillerphoto.com/Moment/Getty Images

In the previous chapter, you learned how to use the capital asset pricing model to estimate the expected return on a company's common stock. If the firm is financed wholly by common stock, then the stockholders own all the firm's assets and are entitled to all the cash flows. In this case, the return required by investors in the common stock equals the company cost of capital.

Most companies, however, are financed by a mixture of securities, including common stock, bonds, and preferred stock. In this case, the company cost of capital is no longer the same as the expected return on the common stock. It depends on the expected return from *all* the securities that the company has issued.

The cost of capital also depends on taxes because interest payments made by a corporation are tax-deductible expenses. Therefore, the company cost of capital is usually calculated as a weighted average of

the *after-tax* cost of debt interest and the "cost of equity"—that is, the expected rate of return investors demand of the firm's common stock. The weights are the fractions of debt and equity in the firm's capital structure. Managers refer to the firm's *weighted-average* cost of capital, or WACC (rhymes with "quack").

Managers use the firm's weighted-average cost of capital to evaluate average-risk investment projects. "Average risk" means that the project's risk matches the risk of the firm's existing assets and operations. This chapter explains how the weighted-average cost of capital is calculated in practice.

Managers calculating WACC can get bogged down in formulas. We want you to understand *why* WACC works, not just how to calculate it. Let's start with "Why?" We'll listen in as a young financial manager struggles to recall the rationale for project discount rates.

13.1 Geothermal's Cost of Capital

Jo Ann Cox, a recent graduate of a prestigious eastern business school, poured a third cup of black coffee and tried again to remember what she once knew about project hurdle rates. Why hadn't she paid more attention in Finance 101? Why had she sold her finance text the day after passing the finance final?

Costas Thermopolis, her boss and CEO of Geothermal Corporation, had told her to prepare a financial evaluation of a proposed expansion of Geothermal's production. She was to report at 9:00 Monday morning. Thermopolis, whose background was geophysics, not finance, not only expected a numerical analysis, but also expected her to explain it to him.

Thermopolis had founded Geothermal in 1996 to produce electricity from geothermal energy trapped deep under Nevada. The company had pioneered this business and had obtained perpetual production rights for a large tract on favorable terms from the U.S. government. When the oil stock in 2007–2008 drove up energy prices worldwide, Geothermal became an exceptionally profitable company. It was currently reporting a rate of return on book assets of 25% per year.

Now, in 2021, energy prices were no longer high and production rights were no longer cheap. The proposed expansion would cost \$30 million and should generate a perpetual after-tax cash flow of \$4.5 million annually. The projected rate of return was $4.5/30 = .15$, or 15%, much less than the profitability of Geothermal's existing assets. However, once the new project was up and running, it would be no riskier than Geothermal's present business.

Jo Ann realized that 15% was not necessarily a bad return—though, of course, 25% would have been better. Fifteen percent might still exceed Geothermal's cost of capital—that is, exceed the expected rate of return that outside investors would demand to invest money in the project. If the cost of capital was less than the 15% expected return, expansion would be a good deal and would generate net value for Geothermal and its stockholders.

Jo Ann remembered how to calculate the cost of capital for companies that used only common stock financing. Briefly, she sketched the argument.

"I need the expected rate of return investors would require from Geothermal's real assets—the wells, pumps, generators, etc.¹ That rate of return depends on the assets' risk. However, the assets aren't traded in the stock market, so I can't observe how risky they have been. I can only observe the risk of Geothermal's common stock.

"But if Geothermal issues only stock—no debt—then shareholders own the entire firm free and clear, with full claim on its assets. In that case, the expected return demanded by investors in the stock must also be the cost of capital for the assets." She jotted down the following identities:

$$\begin{aligned}\text{Value of business} &= \text{value of stock} \\ \text{Risk of business} &= \text{risk of stock}\end{aligned}$$

$$\text{Rate of return on business} = \text{rate of return on stock}$$

Investors' required return from business = investors' required return from stock
If there were no company debt, this would be the right discount rate for Geothermal's expansion plan.

Unfortunately, Geothermal had borrowed a substantial amount of money; its stockholders did not have unencumbered ownership of Geothermal's assets. The expansion project would also justify some extra debt finance. Jo Ann realized that she would have to look at Geothermal's capital structure—its mix of debt and equity financing—and consider the expected rates of return required by debt as well as equity investors.

¹ Investors will invest in the firm's securities only if they offer the same expected return as other equally risky securities. When securities are properly priced, the return that investors can expect from their investments is therefore also the return that they require.

capital structure
The mix of long-term debt
and equity financing.

Assets	Liabilities and Shareholders' Equity			
Market value of assets = value of Geothermal's existing business	\$647	Market value of debt	\$194	(30%)
Total value	\$647	Market value of equity	453	(70%)
		Total value	\$647	(100%)

"Holy Toledo, I've got it!" Jo Ann exclaimed. "If I bought all the securities issued by Geothermal, debt as well as equity, I'd own the entire business. That means ..." She jotted again:

$$\begin{aligned}\text{Value of business} &= \text{value of portfolio of all the firm's} \\ &\quad \text{debt and equity securities} \\ \text{Risk of business} &= \text{risk of portfolio} \\ \text{Rate of return on business} &= \text{rate of return on portfolio} \\ \text{Investor's required return on business} &= \text{investors' required return on} \\ &\quad \text{(company cost of capital) = portfolio}\end{aligned}$$

"All I have to do is calculate the expected rate of return on a portfolio of all the firm's securities. That's easy. The debt's yielding 8%, and Fred, that nerdy banker, says that equity investors want 14%. Suppose he's right. The portfolio would contain 30% debt and 70% equity, so ..."

$$\text{Portfolio return} = (.3 \times 8\%) + (.7 \times 14\%) = 12.2\%$$

It was all coming back to her now. The company cost of capital is just a weighted average of returns on debt and equity, with weights depending on relative market values of the two securities.

"But there's one more thing. Interest is tax-deductible. If Geothermal pays \$1 of interest, taxable income is reduced by \$1, and the firm's tax bill drops by 21 cents (assuming a 21% tax rate). The net cost is only 79 cents. So the after-tax cost of debt is not 8%, but $.79 \times 8 = 6.3\%$.

"Now I can finally calculate the weighted-average cost of capital:

$$\text{WACC} = (.3 \times 6.3\%) + (.7 \times 14\%) = 11.7\%$$

"Looks like the expansion's a good deal. Fifteen's better than 11.7. But I sure need a break."

13.2

The Weighted-Average Cost of Capital

Jo Ann's conclusions were important. It should be obvious by now that the choice of the discount rate can be crucial, especially when the project is long-lived.

Think again what the company cost of capital is, and what it is used for. We define it as the opportunity cost of capital for the firm's existing assets; we use it to value new assets that have the same risk as the old ones. The company cost of capital is the minimum acceptable rate of return when the firm expands by investing in average-risk projects.

company cost of capital
Opportunity cost of capital
for investment in the firm as
a whole. The company cost
of capital is the appropriate
discount rate for an average-
risk investment project
undertaken by the firm.

We first introduced the opportunity cost of capital in Chapter 1. "Opportunity cost" is a shorthand reminder that when the firm invests rather than returning cash to shareholders, the shareholders lose the opportunity to invest in financial markets. If the corporation acts in the shareholders' interests, it will invest their money only if it can find projects that offer higher rates of return than investors could achieve on their own. Therefore, the expected rates of return on investments in financial markets determine the cost of capital for corporate investments.

The company cost of capital is the opportunity cost of capital for the company as a whole. We discussed the company cost of capital in Chapter 12 but did not explain how to measure it when the firm has raised different types of debt and equity financing or how to adjust it for the tax-deductibility of interest payments. The weighted-average cost of capital formula handles these complications.

Calculating Company Cost of Capital as a Weighted Average

When only common stock is outstanding, calculating the company cost of capital is straightforward, though not always easy. For example, a financial manager could estimate beta and calculate shareholders' required rate of return using the capital asset pricing model (CAPM). This would be the expected rate of return investors require on the company's existing assets and operations and also the expected return they will require on new investments that do not change the company's market risk.

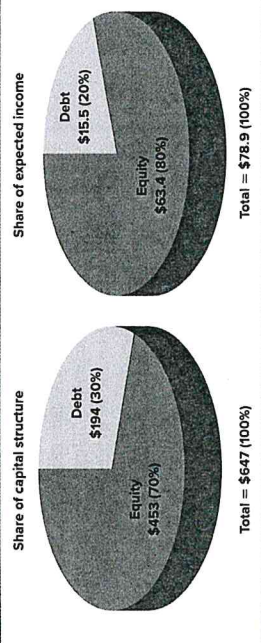
But most companies issue debt as well as equity. **The company cost of capital is a weighted average of the returns demanded by debt and equity investors. The weighted average is the expected rate of return investors would demand on a portfolio of all the firm's outstanding securities.**

Let's review Jo Ann Cox's calculations for Geothermal. To avoid complications, we'll ignore taxes for the next two or three pages. The total market value of Geothermal, which we denote as V , is the sum of the values of the outstanding debt D and the equity E . Thus, firm value is $V = D + E = \$194 \text{ million} + \$453 \text{ million} = \$647 \text{ million}$. Debt accounts for 30% of the value and equity accounts for the remaining 70%. If you held all the shares and all the debt, your investment in Geothermal would be $V = \$647 \text{ million}$. Between them, the debtholders and equityholders own *all* the firm's assets. So V is also the value of these assets—the value of Geothermal's existing business.

Suppose that Geothermal's equity investors require a 14% rate of return on their investment in the stock. What rate of return must a new project provide in order that all investors—both debtholders and stockholders—earn a fair rate of return? The debtholders require a rate of return of $r_{\text{debt}} = 8\%$. So, each year the firm will need to pay interest of $r_{\text{debt}} \times D = .08 \times \$194 \text{ million} = \$15.52 \text{ million}$. The shareholders, who have invested in a riskier security, require an expected return of $r_{\text{equity}} = 14\%$ on their investment of $\$453 \text{ million}$. Thus, in order to keep shareholders happy, the company needs additional income of $r_{\text{equity}} \times E = .14 \times \$453 \text{ million} = \$63.42 \text{ million}$. To satisfy both the debtholders and the shareholders, Geothermal needs to earn $\$15.52 \text{ million} + \$63.42 \text{ million} = \78.94 million . This is equivalent to earning a return of $r_{\text{assets}} = \$78.94/\$647 = .122$, or 12.2%.

Figure 13.1 illustrates the reasoning behind our calculations. The figure shows the amount of income needed to satisfy the debt and equity investors. Notice that debtholders account for 30% of Geothermal's capital structure but receive less than 30% of its expected income. On the other hand, they bear less than a 30% share of risk because they have first cut at the company's income and also first claim on its assets if the company gets in trouble. Shareholders expect a return of more than 70% of Geothermal's income because they bear correspondingly more risk.

FIGURE 13.1 Geothermal's debtholders account for 30% of the company's capital structure, but their share of total income is less than 30% because their 8% return is guaranteed by the company and thus less than the expected return earned by equity investors. Geothermal's stockholders bear more risk and, to compensate, they receive, on average, greater return. Of course, if you buy all the debt and all the equity, you get all the income.



However, if you buy *all* Geothermal's debt and equity, you own its assets lock, stock, and barrel. You receive all the income and bear all the risks. The expected rate of return you'd require on this portfolio of securities is the same return you'd require from unencumbered ownership of the business. This rate of return—12.2%, ignoring taxes—is, therefore, the company cost of capital and the required rate of return from an expansion of the business.

The bottom line (still ignoring taxes) is

Company cost of capital = weighted average of debt and equity returns

The underlying algebra is simple. Debtholders need income of $(r_{\text{debt}} \times D)$, and the equity investors need expected income of $(r_{\text{equity}} \times E)$. The *total* income that is needed is $(r_{\text{debt}} \times D) + (r_{\text{equity}} \times E)$. The amount of their combined existing investment in the company is V . So to calculate the return that is needed on the assets, we simply divide the income by the investment:

$$r_{\text{assets}} = \frac{\text{total income}}{\text{value of investment}} = \frac{(D \times r_{\text{debt}}) + (E \times r_{\text{equity}})}{V} = \left(\frac{D}{V} \times r_{\text{debt}} \right) + \left(\frac{E}{V} \times r_{\text{equity}} \right)$$

For Geothermal,

$$r_{\text{assets}} = (.30 \times 8\%) + (.70 \times 14\%) = 12.2\%$$

This figure is the expected return demanded by investors in the firm's assets.

13.1 Self-Test

Hot Rocks Corp., one of Geothermal's competitors, has issued long-term bonds with a market value of \$50 million and an expected return of 9%. It has 4 million shares outstanding trading for \$10 each. At this price, the shares offer an expected return of 17%. What is the weighted-average cost of capital for Hot Rocks' assets and operations? Assume Hot Rocks pays no taxes.

Use Market Weights, Not Book Weights

The company cost of capital is the expected rate of return that investors demand from the company's assets and operations. The cost of capital must be based on what investors are actually willing to pay for the company's outstanding securities—that is, based on the securities' market values.

Market values usually differ from the values recorded by accountants in the company's books. The book value of Geothermal's equity reflects money raised in the past from shareholders or reinvested by the firm on their behalf. If investors recognize Geothermal's excellent prospects, the market value of equity may be much higher than book, and the debt ratio will be lower when measured in terms of market values rather than book values.

Financial managers use book debt ratios for various purposes, and sometimes they unwittingly look to the book ratios when calculating weights for the company cost of capital. That's a mistake because the company cost of capital measures what investors want from the company and so depends on how they value the company's securities. That value depends on future profits and cash flows, not on accounting history. Book values, while useful for many other purposes, measure only cumulative historical financing; they don't generally measure market values accurately.

13.2 Self-Test

Here is a book balance sheet for Duane S. Burg Associates. Figures are in millions.

Assets		Liabilities and Shareholders' Equity	
Assets (book value)	\$75	Debt	\$25
		Equity	50
	\$75		\$75

Unfortunately, the company has fallen on hard times. The 6 million shares are trading for only \$4 apiece, and the market value of its debt securities is 20% below the face (book) value. Because of the company's large cumulative losses, it will pay no taxes on future income.

Suppose shareholders now demand a 20% expected rate of return. The bonds are now yielding 14%. What is the weighted-average cost of capital?

Taxes and the Weighted-Average Cost of Capital

So far, our examples have ignored taxes. When you calculate a project's NPV, you need to discount the cash flows after tax assuming that the project is wholly equity-financed. That is exactly the approach that we used in Chapter 9, when we valued Blooper's investment in the magnosium mine. Sometimes, you may encounter companies that forecast cash flows before tax and then try to compensate for this by using a higher discount rate. It doesn't work; there is no simple adjustment to the discount rate that will allow you to discount pretax cash flows.

Taxes are also important because most companies are financed by both equity and debt. The interest payments on this debt are deducted from income before tax is calculated. Therefore, the cost to the company is reduced by the amount of this tax saving.

The interest rate on Geothermal's debt is $r_{\text{debt}} = 8\%$. However, with a corporate tax rate of $T_c = .21$, the government bears 21% of the cost of the interest payments. The government doesn't send the firm a check for this amount, but the income tax that the

firm pays is reduced by 21% of its interest expense. Therefore, Geothermal's after-tax cost of debt is only $100 - 21 = 79\%$ of the 8% pretax cost:

$$\begin{aligned}\text{After-tax cost of debt} &= (1 - \text{tax rate}) \times \text{pretax cost} \\ &= (1 - T_c) \times r_{\text{debt}} \\ &= (1 - .21) \times 8\% = 6.3\%\end{aligned}$$

We can now adjust our calculation of Geothermal's cost of capital to recognize the tax savings associated with interest payments:

$$\text{Company cost of capital, after-tax} = (.3 \times 6.3\%) + (.7 \times 14\%) = 11.7\%$$

Now we're back to the weighted-average cost of capital, or WACC. The general formula is

$$\text{WACC} = \left[\frac{D}{V} \times (1 - T_c) r_{\text{debt}} \right] + \left(\frac{E}{V} \times r_{\text{equity}} \right) \quad (13.1)$$

13.3 Self-Test

Criss-Cross Industries has earnings before interest and taxes (EBIT) of \$10 million. Interest payments are \$2 million, and the corporate tax rate is 21%. Construct a simple income statement to show that the debt interest reduces the taxes the firm owes to the government. How much more tax would Criss-Cross pay if it were financed solely by equity?

Example 13.1

Weighted-Average Cost of Capital for Ford

In Chapter 12, Table 12.2, we used the capital asset pricing model to estimate the expected return on the common stock of several companies. We will now use this estimate to compute the weighted-average cost of capital for Campbell Soup Co. in 2020.

Step 1. Calculate capital structure using market value weights. The total market value of Campbell, securities is $V = D + E = \$4,990 + \$13,730 = \$18,720$ million. Debt as a proportion of the total value is $D/V = \$4,990/\$18,720 = .267$, and equity as a proportion of the total is $E/V = \$13,730/\$18,720 = .733$.

Step 2. Determine the required rate of return on each security. In Chapter 12, we estimated that Campbell shareholders required a return of 4.0%. The average yield on Campbell debt was about 2.9%.

Step 3. Calculate a weighted average of the after-tax return on the debt and the return on the equity.² The weighted-average cost of capital is

$$\begin{aligned}\text{WACC} &= \left[\frac{D}{V} \times (1 - T_c) r_{\text{debt}} \right] + \left(\frac{E}{V} \times r_{\text{equity}} \right) \\ &= [.267 \times (1 - .21) \times 2.9\%] + (.733 \times 4.0\%) = 3.5\%\end{aligned}$$

13.4 Self-Test

Calculate WACC for Hot Rocks (Self-Test 13.1) and Burg Associates (Self-Test 13.2), assuming the companies face a 21% corporate income tax rate.

² Financial managers often use "equity" to refer to common stock, even though a firm's equity strictly includes both common and preferred stock. We continue to use r_{equity} to refer specifically to the expected return on the common stock.

What If There Are Three (or More) Sources of Financing?

We have simplified our discussion of the cost of capital by assuming the firm has only two classes of securities: debt and equity. Even if the firm has issued other classes of securities, our general approach to calculating WACC remains unchanged. We simply calculate the weighted-average after-tax return of each security type.

For example, if Campbell had issued preferred stock, we would also need to include this in our calculation of WACC. Preferred stock has some of the characteristics of both common stock and fixed-income securities. Like bonds, preferred stock promises to pay a given, usually level, stream of dividends. Unlike bonds, however, there is no maturity date for the preferred stock. The promised dividends constitute a perpetuity as long as the firm stays in business. Moreover, a failure to come up with the cash to pay the dividends does not push the firm into bankruptcy. Instead, any unpaid dividends simply cumulate; the common stockholders do not receive dividends until the accumulated preferred dividends have been paid. Finally, unlike interest payments, preferred stock dividends are not considered tax-deductible expenses.

How would we calculate WACC for a firm with preferred stock as well as common stock and bonds outstanding? Using P to denote the value of preferred stock, we can generalize Equation 13.1 for WACC as follows:

$$WACC = \left[\frac{P}{V} \times (1 - T_c)r_{\text{deb}} \right] + \left(\frac{P}{V} \times r_{\text{preferred}} \right) + \left(\frac{E}{V} \times r_{\text{equity}} \right) \quad (13.1a)$$

The NPV of Geothermal's Expansion

We now return to Jo Ann Cox and Geothermal's proposed expansion. We want to make sure that both she and you know how to use the weighted-average cost of capital.

Remember that the proposed expansion costs \$30 million and should generate a perpetual cash flow of \$4.5 million per year. A simple cash-flow worksheet might look like this:³

Revenue	\$10.00 million
– Operating expenses	–4.30
= Pretax operating cash flow	5.70
– Tax at 21%	–1.20
After-tax cash flow	\$ 4.50 million

Note that these cash flows do not include the tax benefits of using debt. Geothermal's managers and engineers forecast revenues, costs, and taxes as if the project were to be all-equity-financed. The interest tax shields generated by the project's actual debt financing are not forgotten, however. They are accounted for by using the *after-tax* cost of debt in the weighted-average cost of capital.

Project net present value is calculated by discounting expected cash flow (which is a perpetuity) at Geothermal's 11.7% weighted-average cost of capital:

$$NPV = -30 + \frac{4.5}{.117} = +\$8.46 \text{ million}$$

Expansion will thus add \$8.46 million to the net wealth of Geothermal's owners.

³ For this example we ignore depreciation, a noncash but tax-deductible expense. (If the project is really perpetual, why depreciate?)

Checking Our Logic

Any project offering a rate of return more than 11.7% will have a positive NPV, assuming that the project has the same risk and financing as Geothermal's business. A project offering exactly 11.7% would just break even; it would generate just enough cash to satisfy both debtholders and stockholders.

Let's check that out. Suppose the revenues of the proposed expansion were only just sufficient to generate an NPV of zero. This would require an after-tax cash flow equal to 11.7% of the \$30 million investment, or \$3.51 million. The firm would need \$8.74 million in revenue to generate this cash flow.⁴

Revenue	\$8.74 million
– Operating expenses	–4.30
= Pretax operating cash flow	4.44
– Tax at 21%	–0.93
After-tax cash flow	\$3.51 million

With an investment of \$30 million, the internal rate of return on this perpetuity is indeed 11.7%:

$$\text{Rate of return} = \frac{3.51}{30} = .117, \text{ or } 11.7\%$$

and NPV is exactly zero:

$$NPV = -30 + \frac{3.51}{.117} = 0$$

When we calculated Geothermal's weighted-average cost of capital, we recognized that the company's debt ratio was 30%. When Geothermal's analysts use the weighted-average cost of capital to evaluate the new project, they are *assuming* that the \$30 million additional investment would support the issue of additional debt equal to 30% of the investment, or \$9 million. The remaining \$21 million is provided by the shareholders either in the form of reinvested earnings or through the issue of additional shares.

The following table shows how the cash flows would be shared between the debtholders and shareholders assuming still that the project has zero NPV. We start with the pretax operating cash flow of \$4.44 million:

Cash flow before tax and interest	\$4.44 million
– Interest payment ($0.8 \times \$9 \text{ million}$)	–0.72
= Pretax cash flow	3.72
– Tax at 21%	–0.78
After-tax cash flow	\$2.94 million

Project cash flows before tax and interest are forecast to be \$4.44 million. Out of this figure, Geothermal needs to pay interest of 8% of \$9 million, which comes to \$.72 million. This leaves a pretax cash flow of \$3.72 million, on which the company must pay tax. Taxes equal $.21 \times 3.72 = \$0.78$ million. Shareholders are left with \$2.94 million, just enough to give them the 14% return that they need on their \$21 million investment. (Note that $2.94/21 = .14$, or 14%.) Therefore, everything checks out.

If a project has zero NPV when the expected cash flows are discounted at the weighted-average cost of capital, then the project's cash flows are just sufficient to give debtholders and shareholders the returns they require.

⁴ We assume that the costs of operating the mine remain at \$4.30 million even when revenues are reduced.

13.3

Interpreting the Weighted-Average Cost of Capital

When You Can and Can't Use WACC

Geothermal's weighted-average cost of capital is the rate of return that the firm must expect to earn on its average-risk investments in order to provide an adequate expected return to all its security holders. Strictly speaking, the weighted-average cost of capital is an appropriate discount rate only for a project that is a carbon copy of the firm's existing business. But, often, it is used as a companywide benchmark discount rate; the benchmark may be adjusted upward for unusually risky projects and downward for unusually safe ones.

There is a good musical analogy here. Most of us, lacking perfect pitch, need a well-defined reference point, like middle C, before we can sing on key. But anyone who can carry a tune gets *relative* pitches right. Businesspeople have good intuition about *relative* risks, at least in industries they are used to, but not about absolute risk or required rates of return. Therefore, they set a company- or industrywide cost of capital as a benchmark. This is not the right hurdle rate for everything the company does, but judgmental adjustments can be made for riskier or less risky ventures.

Some Common Mistakes

One danger with the weighted-average formula is that it tempts people to make logical errors. Think back to Jo Ann Cox's estimate of Geothermal's cost of capital:

$$\begin{aligned} \text{WACC} &= \left[\frac{D}{V} \times (1 - T_c) r_{\text{debt}} \right] + \left(\frac{E}{V} \times r_{\text{equity}} \right) \\ &= [.3 \times (1 - .21) 8\%] + (.7 \times 14\%) = 11.7\% \end{aligned}$$

Now suppose that, having read Jo Ann's report, Costas Thermopolis replies, "Aha! I have an idea. Geothermal has a good credit rating. It could easily push up its debt ratio to 50%. If the interest rate is 8% and the required return on equity is 14%, the weighted-average cost of capital would be only:

$$\text{WACC} = [.50 \times (1 - .21) 8\%] + (.50 \times 14\%) = 10.2\%$$

At a discount rate of 10.2%, we can justify a lot more investment."

That reasoning will get the company into trouble. First, if Geothermal increased its borrowing, the lenders would almost certainly demand a higher rate of interest on the debt. Second, as the borrowing increased, the risk of the common stock would also increase and therefore the stockholders would demand a higher return.

There are actually two costs of debt finance. The explicit cost of debt is the rate of interest that bondholders demand. But there is an additional implicit cost because borrowing increases the required return to equity. When Costas Thermopolis jumped to the conclusion that Geothermal could lower its weighted-average cost of capital to 10.2% by borrowing more, he was recognizing only the explicit cost of debt and not the implicit cost.

13.5 Self-Test

Costas Thermopolis now points out that Geothermal proposes to finance its expansion entirely by borrowing at an interest rate of 8%. He argues that this is, therefore, the appropriate discount rate for the project's cash flows. Is he right?

How Changing Capital Structure Affects Expected Returns

We will illustrate how changes in capital structure affect expected returns by focusing on the simplest possible case, where the corporate tax rate T_c is zero. We saw earlier that in this case Geothermal's cost of capital was 12.2%.

Think about what would happen if Geothermal borrows an additional \$100 million and uses the cash to buy back and retire \$100 million of its common stock. If there are no corporate taxes, the change in capital structure does not affect the total cash that Geothermal pays out to its security holders, and it does not affect the risk of those cash flows. Therefore, if investors require a return of 12.2% on the total package of debt and equity before the financing, they must require the same 12.2% return on the package afterward. The weighted-average cost of capital is therefore unaffected by the change in the capital structure.

Is that surprising? After all, the required return on debt is lower than the required return on equity, so you might expect the additional borrowing to *reduce* the weighted average cost of capital. The reason that it does not do so is that the return on the individual securities changes. Because the company has more debt than before, the debt is riskier, and debtholders are likely to demand a higher return. Increasing the amount of debt also makes the equity riskier and increases the return that shareholders require. We will return to this point in Chapter 16.

What Happens When the Corporate Tax Rate Is Not Zero

We have shown that when there are no corporate taxes, the weighted-average cost of capital is unaffected by a change in capital structure. Unfortunately, taxes can complicate the picture.³ For the moment, just remember:

- The weighted-average cost of capital is the right discount rate for average risk capital investments.
- The weighted-average cost of capital is the return the company needs to earn after tax in order to satisfy all its security holders.
- If the firm increases its debt ratio, both the debt and the equity will become riskier. The debtholders and equityholders require a higher return to compensate for increased risk.

13.4

Practical Problems: Measuring Capital Structure

We have explained the formula for calculating the weighted-average cost of capital. We will now look at some of the practical problems in applying that formula. We will give Jo Ann Cox a break, and suppose that you have been asked to estimate Big Oil's weighted-average cost of capital. Your first step is to work out Big Oil's capital structure. But where do you get the data?

Financial managers usually start with the company's accounts, which show the book value of debt and equity, whereas the weighted-average cost of capital formula calls for their *market* values. A little work and a dash of judgment are needed to go from one to the other.

Table 13.1 shows the debt and equity issued by Big Oil. The firm has borrowed \$200 million from banks and has issued a further \$200 million of long-term bonds. These bonds have a coupon rate of 8% and mature at the end of 12 years. Finally, there are 100 million shares of common stock outstanding, each with a par value of \$1.

³ There's nothing wrong with our formulas and examples, provided that the tax deductibility of interest payments doesn't change the aggregate risk of the debt and equity investors. However, if the tax savings from deducting interest are treated as safe cash flows, the formulas get more complicated. If you really want to dive into the tax-adjusted formulas showing how WACC changes with capital structure, we suggest Chapter 19 in R. A. Brealey, S. C. Myers, F. Allen, and A. Edmans, *Principles of Corporate Finance*, 14th ed. (New York: McGraw Hill LLC, 2022).

TABLE 13.1 Big Oil's book (accounting) balance sheet (dollar values in millions).

Assets		Long-Term Liabilities and Equity	
Net working capital (= current assets - current liabilities)		Bank debt	\$200
Property, plant, and equipment (PP&E)	\$120	Long-term bonds (12-year maturity, 8% coupon)	200
Other long-term assets	620	Common stock (100 million shares, par value = \$1)	100
		Retained earnings	300
			\$800
			100.0%

But the accounts also recognize that Big Oil has in past years plowed back into the firm \$300 million of retained earnings. The total book value of the equity shown in the accounts is \$100 million + \$300 million = \$400 million.

Notice that the right side of Table 13.1 contains long-term financing. Current liabilities have been subtracted on the left as an offset to current assets. Thus, *net working capital* equals current assets minus current liabilities. We set up the balance sheet in this way in order to compute the weighted-average cost of capital, which is normally defined as the combined cost of *long-term* debt financing and equity.⁶ Thus, the balance sheet shows only sources of long-term financing on the right.

The figures shown in Table 13.1 are taken from Big Oil's annual accounts and are, therefore, book values. Sometimes the differences between book values and market values are negligible. For example, consider the \$200 million that Big Oil owes the bank. The interest rate on bank loans is usually linked to the general level of interest rates. Thus, if interest rates rise, the rate charged on Big Oil's loan also rises to maintain the loan's value. As long as Big Oil is reasonably sure to repay the loan, it is worth close to \$200 million. Most financial managers most of the time are willing to accept the book value of bank debt as a fair approximation of its market value.

What about Big Oil's long-term bonds? Since the bonds were originally issued, long-term interest rates have risen to 9%.⁷ We can calculate the value today of each bond as follows:⁸ There are 12 coupon payments of $.08 \times 200 = \$16$ million and then repayment of face value 12 years out. Thus, the final cash payment to the bondholders is \$216 million. All the bond's cash flows are discounted back at the *current* interest rate of 9%:

$$PV = \frac{16}{1.09} + \frac{16}{(1.09)^2} + \frac{16}{(1.09)^3} + \dots + \frac{216}{(1.09)^{12}} = \$185.7$$

Therefore, the bonds are worth only \$185.7 million, 93% of their face value.

If you used the book value of Big Oil's long-term debt rather than its market value, you would be a little bit off in your calculation of the weighted-average cost of capital, but probably not seriously so.

The really big errors are likely to arise if you use the book value of equity rather than its market value. The \$400 million book value of Big Oil's equity measures the total amount of cash that the firm has raised from shareholders in the past or has retained and invested on their behalf. But perhaps Big Oil has been able to find projects that were worth more than they originally cost, or perhaps the value of the assets

⁶ Sometimes, companies finance long-term investment with short-term debt, rolling over the short-term debt as it matures. In this case, the permanent component of short-term debt could remain on the right side of the balance sheet, and the cost of short-term debt could be included in WACC. We assume that Big Oil's bank debt is part of its long-term financing.

⁷ If Big Oil's bonds are traded, you can simply look up their price. (You can find prices for bond trades on www.finnn.org/marketdata.) But many bonds are not regularly traded, and in such cases, you need to infer their price by calculating the bond's value using the rate of interest offered by similar bonds.

⁸ We assume that coupon payments are annual. Most bonds in the United States actually pay interest twice a year.

TABLE 13.2 Big Oil's market-value balance sheet (dollar values in millions).

Assets		Liabilities and Equity	
Net working capital (= current assets - current liabilities)		Bank debt	\$ 200.0
Value of PP&E and other long-term assets (including intangible assets)	1,465.7	Long-term bonds (market value = 93% par value)	185.7
		Total debt	385.7
		Common stock (100 million shares, par value = \$1)	1,200.0
Value		Value	\$1,585.7
			100.0%

has increased with inflation. Perhaps investors see great future investment opportunities for the company. All these considerations determine what investors are willing to pay for Big Oil's common stock.

Big Oil's stock price is \$12 a share. Thus, the total *market value* of the stock is

Number of shares $\times$ share price = 100 million $\times$ \$12 = \$1,200 million

In Table 13.2, we show a *market-value* balance sheet for Big Oil. You can see that debt accounts for 24.3% of company value ($D/V = .243$) and equity accounts for 75.7% ($E/V = .757$). These are the proportions to use when calculating the weighted-average cost of capital. Notice that if you looked only at the book values shown in Table 13.1, you would mistakenly conclude that debt and equity each accounted for 50% of value.

13.6 Self-Test

Here is the capital structure shown in Executive Fruit's book balance sheet:

Debt	\$4.1 million	45.0%
Preferred stock	2.2	24.2
Common stock	2.8	30.8
Total	\$9.1 million	100.0%

Explain why the percentage weights given here should not be used in calculating Executive Fruit's WACC.

More Practical Problems: Estimating Expected Returns

To calculate Big Oil's weighted-average cost of capital, you also need the rate of return that investors require from each security.

The Expected Return on Bonds

We know that Big Oil's bonds offer a yield to maturity of 9%. As long as the company does not go belly-up, that is the rate of return investors can expect to earn from holding Big Oil's bonds. If there is any chance that the firm may be unable to repay the debt, however, the yield to maturity of 9% represents the most favorable outcome and the expected return is lower than 9%.

For most large and healthy firms, the probability of bankruptcy is sufficiently low that financial managers are content to take the promised yield to maturity on the bonds as a measure of the expected return. But this approach doesn't always work. For

example, at the height of the financial crisis in 2008, the yield to maturity on Ford's bonds rose to about 20%. No one thought that this was the expected rate of return on the bonds. Rather, the market recognized that there was a chance that Ford might not be able to come up with all the cash flows promised to its bond investors. The yield based on expected cash flows was far lower than the yield based on promised cash flows.

The Expected Return on Common Stock

Estimates Based on the Capital Asset Pricing Model In the previous chapter, we showed you how to use the capital asset pricing model to estimate the expected rate of return on common stock. The capital asset pricing model tells us that investors demand a higher rate of return from stocks with high betas. The formula is

$$\text{Expected return on stock} = \text{risk-free interest rate} + \left(\text{stock's beta} \times \text{expected market risk premium} \right)$$

Financial managers and economists measure the risk-free rate of interest by the yield on Treasury debt securities. To measure the expected market risk premium, they usually look back at capital market history, which suggests that investors have received about an extra 7% a year from investing in common stocks rather than Treasury bills.⁹ Yet, wise financial managers use this evidence with considerable humility, for who is to say whether investors in the past received more or less than they expected or whether investors today require a higher or lower reward for risk than their parents did?

Let's suppose Big Oil's common stock beta is estimated at .85, the risk-free interest rate (r_f) at the time is 6%, and the expected market risk premium ($r_m - r_f$) is 7%. Then the CAPM would put Big Oil's cost of equity at

$$\begin{aligned} \text{Cost of equity} &= r_{\text{equity}} = r_f + \beta(r_m - r_f) \\ &= 6\% + .85(7\%) = 12\% \end{aligned}$$

Estimates Based on the Dividend Discount Model Whenever you are given an estimate of the expected return on a common stock, always look for ways to check whether it is reasonable. One check on the estimates provided by the CAPM can be obtained from the dividend discount model (DDM). In Chapter 7, we showed you how to use the constant-growth DDM formula to estimate the return that investors expect from different common stocks. Remember the formula: If dividends are expected to grow indefinitely at a constant rate g , then the price of the stock is equal to

$$P_0 = \frac{\text{DIV}_1}{r_{\text{equity}} - g}$$

where P_0 is the current stock price, DIV_1 is the forecast dividend at the end of the year, and r_{equity} is the expected return from the stock. We can rearrange this formula to provide an estimate of r_{equity} :

$$r_{\text{equity}} = \frac{\text{DIV}_1}{P_0} + g \quad (13.2)$$

In other words, the expected return on equity is equal to the dividend yield (DIV_1/P_0) plus the expected perpetual growth rate in dividends (g).

⁹ The Treasury bill rate is the customary measure of the risk-free rate of interest. There is a mismatch, however, in using a short-term bill rate to calculate the cost of equity in a WACC. The WACC is used to discount cash flows that may arrive many years in the future. Discounting a distant future cash flow at a short-term rate doesn't make sense, especially when monetary policy forces the short-term rate down close to zero. Therefore, financial managers calculating WACC usually estimate the cost of equity using a long-term Treasury bond rate. In this case, the market risk premium must be defined as the expected difference between the returns on the stock market and on long-term Treasury bonds. Notice in Table 11.1 that the average return on Treasury bonds has been 1.7 percentage points above the average return on bills. Suppose a financial manager uses a long-term bond yield to estimate the cost of equity. If the manager relies on history, he or she would use a market risk premium that is 1.7 percentage points lower than the premium versus bills.

This constant-growth dividend discount model is widely used in estimating expected rates of return on common stocks of public utilities. Utility stocks have a fairly stable growth pattern and are therefore tailor-made for the constant-growth formula.

Remember that the constant-growth formula will get you into trouble if you apply it to firms with very high current rates of growth. Such growth cannot be sustained indefinitely. Using the formula in these circumstances will lead to an overestimate of the expected return.

Beware of False Precision Do not expect estimates of the cost of equity to be precise. In practice, you can't know whether the capital asset pricing model fully explains expected returns or whether the assumptions of the dividend discount model hold exactly. Even if your formulas were right, the required inputs would be noisy and subject to error. Thus, a financial analyst who can confidently locate the cost of equity in a band of two or three percentage points is doing pretty well. In this endeavor, it is perfectly okay to conclude that the cost of equity is, say, "about 15%" or "somewhere in a range from 14% to 16%."¹⁰

Sometimes, accuracy can be improved by estimating the cost of equity or WACC for an industry or a group of comparable companies. This cuts down the "noise" that plagues single-company estimates. Suppose, for example, that you are able to identify three companies with investments and operations similar to Big Oil's. The average WACC for these three companies would be a valuable check on your estimate of WACC for Big Oil alone.

Or suppose that Big Oil is contemplating investment in oil refining. For this venture, Big Oil's existing WACC is probably not right; it needs a discount rate reflecting the risks of the refining business. It could therefore try to estimate WACC for a sample of oil refining companies. If too few "pure-play" refining companies were available—most oil companies invest in production and marketing as well as refining—an industry WACC for a sample of large oil companies could be a useful check or benchmark.

13.7 Self-Test

You have identified a group of oil companies with similar businesses and capital structures to Big Oil. The average estimated beta of their common stocks is 1.1. Use this "industry beta" to reestimate the expected rate of return on Big Oil's common stock, assuming, of course, that the CAPM is true. Recalculate Big Oil's weighted-average cost of capital.

The Expected Return on Preferred Stock

Preferred stock that pays a fixed annual dividend can be valued from the perpetuity formula:

$$\text{Price of preferred} = \frac{\text{dividend}}{r_{\text{preferred}}}$$

where $r_{\text{preferred}}$ is the appropriate discount rate for the preferred stock. Therefore, we can infer the required rate of return on preferred stock by rearranging the valuation formula to

$$r_{\text{preferred}} = \frac{\text{dividend}}{\text{price of preferred}} \quad (13.3)$$

¹⁰ The calculations in this chapter have been done to one or two decimal places just to avoid confusion from rounding.

TABLE 13.3 Data needed to calculate Big Oil's weighted-average cost of capital (dollar values in millions).

Security Type	Capital Structure	Required Rate of Return
Debt	$D = \$385.7$	$DIV = 0.243$
Common stock	$E = \$1,200.0$	$EIV = 0.757$
Total	$V = \$1,585.7$	

Note: Corporate tax rate = $T_c = 21$. We assume that the interest rate on Big Oil's bank debt is the same as on its bonds.

For example, if a share of preferred stock sells for \$20 and pays a dividend of \$2 per share, the expected return on preferred stock is $r_{\text{preferred}} = \$2/\$20 = 10\%$, which is simply the dividend yield.

Adding It All Up

Once you have worked out Big Oil's capital structure and estimated the expected return on its securities, you require only simple arithmetic to calculate the weighted-average cost of capital. Table 13.3 summarizes the necessary data. Now all you need to do is plug the data in Table 13.3 into the weighted-average cost of capital formula:

$$WACC = \left[\frac{D}{V} \times (1 - T_c) r_{\text{deb}} \right] + \left(\frac{E}{V} \times r_{\text{equity}} \right) \\ = [243 \times (1 - 21\%) / 1585.7] + (757 \times 12\%) = 10.8\%$$

Suppose that Big Oil needs to evaluate a project with the same risk as its existing business. If the project would also support a 24.3% debt ratio, the 10.8% weighted-average cost of capital is the appropriate discount rate for the cash flows.

Real-Company WACCs

Big Oil is entirely hypothetical. Therefore, you might be interested in looking at Table 13.4, which gives some estimates of the weighted-average cost of capital for a sample of real companies. As you do so, remember that any estimate of the cost of capital for a single company can be way off the true cost. You should always try to check your estimate by looking at the cost of capital for a group of similar companies.

Ticker	Company	Beta	Expected Return on equity (%)	Interest Rate on debt (%)	Proportion of Equity (E/V)	Proportion of Debt (D/V)	WACC (%)
X	U.S. Steel	3.03	22.2	5.7	0.34	0.66	10.5
MRO	Marathon Oil	2.35	17.4	2.3	0.40	0.60	8.1
AMZN	Amazon	1.51	11.5	1.5	0.96	0.04	11.1
IBM	IBM	1.33	10.3	1.6	0.65	0.35	7.2
BA	Boeing	1.19	9.3	2.9	0.81	0.19	8.0
F	Ford	1.09	8.6	3.5	0.22	0.78	4.1
UNP	Union Pacific	1.07	8.5	1.9	0.84	0.16	7.4
GOOG	Google/Alphabet	1.01	8.1	1.5	0.99	0.01	8.0
DIS	Disney	1.00	8.0	1.9	0.86	0.14	7.1
XOM	ExxonMobil	1.00	8.0	1.5	0.84	0.16	6.9
INTC	Intel	0.91	7.3	1.6	0.89	0.11	6.7
PFE	Pfizer	0.65	5.6	1.5	0.85	0.15	4.9
PCG	Pacific Gas & Electric	0.55	4.9	2.9	0.45	0.55	3.5
SBUX	Starbucks	0.51	4.5	2.1	0.90	0.10	4.3
MCD	McDonald's	0.45	4.1	2.1	0.78	0.22	3.6
CPB	Campbell Soup	0.43	4.0	2.9	0.73	0.27	3.5
KO	Coca-Cola	0.43	4.0	1.6	0.88	0.12	3.7
WMT	Walmart	0.35	3.5	1.5	0.86	0.14	3.1
NEM	Newmont Mining	-0.01	0.9	2.3	0.88	0.12	1.0

Note:

1. Expected return on equity = $1\% + \text{Beta} \times 7\%$.
2. Interest rate on debt is estimated as yield to maturity in September 2020 on comparably rated corporate bonds (with ratings found on the FINRA website).
3. D is the book value of the firm's debt, and E is the market value of equity.
4. $WACC = (DIV) \times (1 - 21\%) \times r_{\text{deb}} + (EIV) \times r_{\text{equity}}$.

13.6 Valuing Entire Businesses

Companies frequently buy and sell entire businesses. Do the methods that we have used in this chapter to value new projects also work for entire businesses?

Sure! As long as the company's debt ratio is expected to remain fairly constant, you can treat the company as one big project and discount its cash flows by the weighted-average cost of capital.¹¹ The result is the combined value of the company's debt and equity. If you want to know just the value of the equity, you must remember to subtract the value of the debt from the company's total value.

Suppose that you are interested in buying Establishment Industry's deconstruction division. The problem is to figure out what it is worth. Table 13.5 sets out your forecasts for the next six years. Row 8 shows the expected cash flow from operations. This is equal to the expected profit after tax plus depreciation. Remember, depreciation is not a cash outflow, and therefore you need to add it back when calculating the operating cash flow. Row 9 in the table shows the forecast investment in fixed assets and net working capital.

The operating cash flow *less* investment expenditures is the amount of cash that the business can pay out to investors after paying for all investments necessary for growth. This is the deconstruction division's **free cash flow (FCF)** (row 10 in the table). Notice that the free cash flow is negative in the early years. The firm is running a cash deficit not because it is unprofitable but because it is growing so fast and investing so much in the business.

The forecast cash flows in Table 13.5 do not include a deduction for debt interest. But we will not forget that acquisition of the deconstruction business will support additional debt. We will recognize this by discounting the free cash flows by the weighted-average cost of capital, which reflects both the firm's capital structure and the tax deductibility of its interest payments.

Suppose that a sensible capital structure for the deconstruction operation is 60% equity and 40% debt.¹² You estimate that the required rate of return on the equity is 12% and that the business could borrow at an interest rate of 5%. The weighted-average cost of capital is therefore

$$WACC = \left[\frac{D}{V} \times (1 - T_c) r_{\text{deb}} \right] + \left(\frac{E}{V} \times r_{\text{equity}} \right) \\ = [4 \times (1 - 21\%) 5\%] + (6 \times 12\%) = 8.78\%$$

TABLE 13.5 Forecasts of operating cash flow and investment for the deconstruction division (thousands of dollars). Rapid expansion means that free cash flow is negative in the early years because investment outstrips the cash flow from operations. Free cash flow turns positive when growth slows down.

	1	2	3	4	5	6
1. Sales	\$1,400.0	\$1,680.0	\$2,016.0	\$2,318.4	\$2,666.2	\$2,932.8
2. Costs	1,190.0	1,428.0	1,713.6	1,970.6	2,266.2	2,492.9
3. EBITDA = 1 - 2	210.0	252.0	302.4	347.8	399.9	439.9
4. Depreciation	100.8	121.0	145.2	166.9	160.0	140.8
5. Profit before tax = 3 - 4	109.2	131.0	157.2	180.8	240.0	299.1
6. Tax at 21%	22.9	27.5	33.0	38.0	50.4	62.8
7. Profit after tax = 5 - 6	86.3	103.5	124.2	142.9	189.6	236.3
8. Operating cash flow = 4 + 7	187.1	224.5	269.4	309.8	349.5	377.1
9. Investment in fixed assets and net working capital	180.0	289.0	346.8	348.4	102.0	-19.2
10. Free cash flow = 8 - 9	7.1	-64.5	-77.4	-38.6	247.5	396.3

*EBITDA = earnings before interest, taxes, depreciation, and amortization.

¹¹ In Chapter 7, we showed how to value a business that is financed entirely by equity. In this section, we extend that discussion and show how to deal with a business that is partly financed by debt.

Calculating the Value of the Deconstruction Business

The value of the deconstruction operation is equal to the discounted value of the free cash flows (FCFs) out to a horizon year plus the forecast value of the business at the horizon, also discounted back to the present. That is,

$$PV = \frac{FCF_1}{1 + WACC} + \frac{FCF_2}{(1 + WACC)^2} + \dots + \frac{FCF_H}{(1 + WACC)^H} + \frac{PV_H}{(1 + WACC)^H}$$

$\underbrace{\hspace{10em}}_{PV(\text{free cash flows})} \quad + \quad \underbrace{\hspace{5em}}_{PV(\text{horizon value})}$

PV_H stands in for the value of free cash flows in periods $H + 1, H + 2$, and so on. We assume that beginning in year 6 the business is expected to settle down to steady growth of 5% a year, and so we have picked year 5 as the horizon year.

We saw in Chapter 7 that there are several common formulas or rules of thumb for estimating horizon value. Here we will just use the constant-growth formula that we introduced in that chapter:

$$\text{Horizon value}_5 = \frac{\text{free cash flow in year 6}}{r - g} = \frac{396.3}{.0878 - .05} = \$10,484 \text{ thousand}$$

We now have all we need to calculate the value of the deconstruction business today. We add up the present values of the free cash flows in the first five years and that of the horizon value:

$$\begin{aligned} PV(\text{business}) &= PV(\text{free cash flows years 1-5}) + PV(\text{horizon value}) \\ &= \frac{7.1}{1.0878} + \frac{-64.5}{(1.0878)^2} + \frac{-77.4}{(1.0878)^3} + \frac{-38.6}{(1.0878)^4} + \frac{247.5}{(1.0878)^5} + \frac{10,484}{(1.0878)^5} \\ &= \$6,910 \text{ thousand} \end{aligned}$$

Notice that when we use the weighted-average cost of capital to value a company, we are asking, "What is the combined value of the company's debt and equity?" If you need to value the equity, you must subtract the value of any outstanding debt. Suppose that the deconstruction business is partly financed with \$2,764,000 of debt, 40% of the overall value of \$6,910,000. Then the equity in the business is worth only \$6,910,000 - \$2,764,000 = \$4,146,000.

13.8 Self-Test

Managers often use rules of thumb to check their estimates of horizon value. Suppose you observe that the value of the debt plus equity of a typical mature deconstruction business is 15 times its EBITDA. (EBITDA is defined at line 3 of Table 13.5.) If your operation sold in year 5 at a similar multiple of EBITDA, how would your estimate of the present value of the operation change?

¹² By this we mean that it makes sense to finance 40% of the present value of the business by debt. Remember that we use market-value weights to compute WACC. Debt as a proportion of book value may be more or less than 40%.

SUMMARY

How do firms compute weighted-average costs of capital? (LO13-1)

- Here's the WACC formula one more time:

$$WACC = \left[\frac{D}{V} \times (1 - T_c) r_{\text{debt}} + \frac{E}{V} \times r_{\text{equity}} \right]$$

- The WACC is the expected rate of return on the portfolio of debt and equity securities issued by the firm.
- The required rate of return on each security is weighted by its proportion of the firm's total market value (not book value).
- Because interest payments reduce the firm's income tax bill, the required rate of return on debt is measured after tax, as $(1 - T_c) r_{\text{debt}}$.

Why do firms compute weighted-average costs of capital? (LO13-2)

- The WACC provides a standard discount rate for average-risk projects. An "average-risk" project is one that has the same risk as the firm's existing operations and that supports the same relative amount of debt.

What about projects that are not average? (LO13-2)

- The weighted-average cost of capital can still be used as a benchmark. The benchmark may be adjusted up for unusually risky projects and down for unusually safe ones.

What happens when capital structure changes? (LO13-2)

- Increasing the debt ratio will increase the risk borne by both debt and equity investors and cause them to demand higher returns.
- However, this does not necessarily mean that the overall WACC will increase because more weight is put on the cost of debt, which is less than the cost of equity.
- If we ignore taxes, the overall cost of capital remains constant as the fractions of debt and equity change.

How do firms measure capital structure? (LO13-3)

- To measure capital structure the firm needs to measure the market value of each source of financing as a proportion of total market value.
- Although the WACC formula is usually written assuming the firm's capital structure includes just two classes of securities, debt and equity, if there is another class, say preferred stock, the formula must expand to include it. In this case, we would estimate $r_{\text{preferred}}$, the rate of return demanded by preferred stockholders; determine P/V , the fraction of market value accounted for by preferred; and add $r_{\text{preferred}} \times P/V$ to the equation.
- Of course, the weights in the WACC formula always add up to 1. If the third class of securities is preferred stock, then the weights on each class would satisfy $D/V + P/V + E/V = 1$.

How are the costs of debt and equity calculated? (LO13-4)

- The cost of debt (r_{debt}) is the expected return demanded by debtholders. For healthy firms this is the interest rate that the company would pay on new debt issued to finance its investment projects.
- The cost of preferred ($r_{\text{preferred}}$) is the preferred dividend divided by the market price of a preferred share.
- The hardest part is estimating the cost of equity (r_{equity}), the expected rate of return on the firm's shares. Financial managers commonly use the capital asset pricing model to estimate expected return. But for mature, steady-growth companies, it can also make sense to use the constant-growth dividend discount model.
- Remember, estimates of expected return are less reliable for a single firm's stock than for a sample of comparable-risk firms. Therefore, managers also consider WACCs calculated for industries.

Can WACC be used to value an entire business? (LO13-5)

- You can think of the business as a very large project. Forecast the business's operating cash flows (after-tax profits plus depreciation), and subtract the future investments in plant and equipment and in net working capital. The resulting **free cash flows** can then be discounted back to the present at the weighted-average cost of capital.
- Of course, the cash flows from a company may stretch far into the future. Financial managers therefore typically produce detailed cash flows only up to some horizon date and then estimate the remaining value of the business at the horizon.

LISTING OF EQUATIONS

$$13.1 \quad WACC = \left[\frac{D}{V} \times (1 - T_c) r_{deb} \right] + \left(\frac{E}{V} \times r_{equity} \right)$$

$$13.1a \quad WACC = \left[\frac{D}{V} \times (1 - T_c) r_{deb} \right] + \left(\frac{P}{V} \times r_{preferred} \right) + \left(\frac{E}{V} \times r_{equity} \right)$$

$$13.2 \quad r_{equity} = \frac{DIV_1}{P_0} + g$$

$$13.3 \quad r_{preferred} = \frac{\text{dividend}}{\text{price of preferred}}$$

QUESTIONS AND PROBLEMS



- Changes in Capital Structure. Look at our calculation of Big Oil's WACC in Section 13.5. (LO13-1)

- Suppose Big Oil is excused from paying taxes. What would be its WACC?
- Now suppose that, after the tax rate has fallen to zero, Big Oil makes a large stock issue and uses the proceeds to pay off all its debt. What would be the cost of equity after the issue?

- WACC. Here is some information about Stokenchurch Inc.:

Beta of common stock = 1.2
 Treasury bill rate = 4%
 Market risk premium = 7.5%
 Yield to maturity on long-term debt = 6%
 Book value of equity = \$440 million
 Market value of equity = \$880 million
 Long-term debt outstanding = \$880 million
 Corporate tax rate = 21%

What is the company's WACC? (LO13-1)

- WACC, Reactive Power Generation has the following capital structure. Its corporate tax rate is 21%. What is its WACC? (LO13-1)

Security	Market Value	Required Rate of Return
Debt	\$20 million	6%
Preferred stock	10 million	8
Common stock	50 million	12

- WACC. The common stock of Buildwell Conservation & Construction Inc. (BCCI) has a beta of .9. The Treasury bill rate is 4%, and the market risk premium is estimated at 8%. BCCI's capital structure is 30% debt, paying a 5% interest rate, and 70% equity. Buildwell pays tax at 21%. (LO13-1)

- What is BCCI's cost of equity capital?
 - What is its WACC?
- If BCCI is presented with a normal project with an internal rate of return of 12%, should it accept the project?
- Calculating WACC. The total book value of WTC's equity is \$10 million, and book value per share is \$20. The stock has a market-to-book ratio of 1.5, and the cost of equity is 15%. The firm's bonds have a face value of \$5 million and sell at a price of 110% of face value. The yield to maturity on the bonds is 9%, and the firm's tax rate is 21%. Find the company's WACC. (LO13-1)
 - WACC, Nodet Inc. is a firm with all-equity financing. Its equity beta is .80. The Treasury bill rate is 4%, and the market risk premium is expected to be 10%. (LO13-1)
 - What is Nodet's asset beta?
 - What is Nodet's WACC?
 - WACC. Look at the following book-value balance sheet for University Products Inc. The preferred stock currently sells for \$15 per share and pays a dividend of \$2 a share. The common stock sells for \$20 per share and has a beta of .8. There are 1 million common shares outstanding. The market risk premium is 10%, the risk-free rate is 6%, and the firm's tax rate is 21%. (LO13-1)

BOOK-VALUE BALANCE SHEET (Figures in \$ millions)			
Assets		Liabilities and Net Worth	
Cash and short-term securities	\$ 1	Bonds, coupon = 8%, paid annually (maturity = 10 years, current yield to maturity = 9%)	\$10
Accounts receivable	3	Preferred stock (par value \$20 per share)	2
Inventories	7	Common stock (par value \$.10)	0.1
Plant and equipment	21	Additional paid-in stockholders' equity	9.9
Total	\$32	Retained earnings	10
			\$32

- What is the market debt-to-value ratio of the firm?
 - What is University's WACC?
- Changes in Capital Structure. Look back at Section 13.5. Suppose Big Oil is excused from paying taxes. It starts from the financing mix in Table 13.3, and then borrows an additional \$200 million from the bank. It uses this cash to pay out a special \$200 million dividend, leaving its assets and operations unchanged. (LO13-1)
 - What happens to Big Oil's WACC, still assuming it pays no taxes?
 - What happens to its cost of equity?
 - WACC. True or false? (LO13-2)
 - If a company uses its company cost of capital to value all projects, it will overestimate the value of high-risk projects.
 - It is okay to use the company cost of capital to value every project as long as the high risks of some projects are offset by the low risks of others.
 - A company can reduce the discount rate for a project by financing it with debt.
 - Company versus Project Discount Rates. Geothermal's WACC is 11.7%. Executive Fruit's WACC is 12.3%. Now Executive Fruit is considering an investment in geothermal power production. (LO13-2)
 - Should it discount project cash flows at 12.3%?
 - What would be a better discount rate for this investment?
 - Project Discount Rate. The total market value of the equity of Okefenokee Condos is \$6 million, and the total value of its debt is \$4 million. The treasurer estimates that the beta of the stock currently is 1.2 and that the expected risk premium on the market is 10%. The Treasury bill rate is 4%, and investors believe that Okefenokee's debt is essentially free of default risk. (LO13-2)
 - What is the required rate of return on Okefenokee stock?
 - Estimate the WACC assuming a tax rate of 21%.